Schur's Lemma and Scalar Endomorphisms

An End-to-End Meta-Math Walkthrough

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Abstract

Equivariance turns kernel and image into invariant subspaces. Algebraic closure then produces a scalar that makes an endomorphism singular, forcing it to vanish after subtraction. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RT-02 / Representation Theory and Lie Theory / Report v1

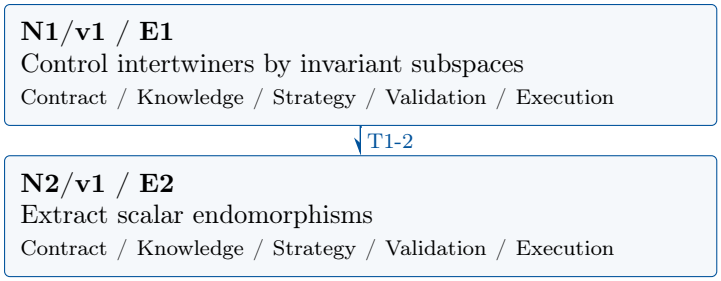
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let V, W be nonzero irreducible representations of a group G over a field k. Every nonzero intertwiner $T : V \to W$ is an isomorphism. If k is algebraically closed and V is finite-dimensional, every equivariant endomorphism of V is scalar. Consequently for finite-dimensional irreducibles over such k, $\dim_k \mathrm{Hom}_G(V, W)$ is one when $V \cong W$ and zero otherwise.*

A. Foundations and permitted imports

Use kernels and images of linear maps, eigenvalue existence for a linear operator on a nonzero finite-dimensional vector space over an algebraically closed field, and linear dimension. The first assertion does not require finite dimension or algebraic closure.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. Every nonzero G-intertwiner $T : V \to W$ is an equivariant isomorphism.

Receiving activity. N2.1

Compatibility. E1/v1 and a finite-dimensional irreducible V over algebraically closed k. Check: Eigenvalue existence is precisely where algebraic closure and finite dimension enter.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Control intertwiners by invariant subspaces

Object and role. Apply irreducibility separately to the kernel and image. **Local obligation:** Every nonzero G-intertwiner $T : V \to W$ is an equivariant isomorphism.

Inputs. $T\rho_V(g) = \rho_W(g)T$ for every g, and irreducible V, W.

C — Contract

Publish E1/v1 only after establishing this obligation: Every nonzero G-intertwiner $T : V \to W$ is an equivariant isomorphism.

K — Knowledge

$T\rho_V(g) = \rho_W(g)T$ for every g, and irreducible V, W. Retain the stated assumptions and import versions.

S — Strategy

Apply irreducibility separately to the kernel and image.

V — Validation

Nonzero T excludes the full kernel and zero image; check both injectivity and surjectivity.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 If $v \in \ker T$, then $T\rho_V(g)v = \rho_W(g)Tv = 0$, so the kernel is invariant. The image is invariant because $\rho_W(g)Tv = T\rho_V(g)v$.

N1.2 Irreducibility gives $\ker T = 0$ or V and $\operatorname{im} T = 0$ or W. Nonzero T forces $\ker T = 0$ and image W. Thus T is a bijective linear map, and multiplying the intertwining equality by its inverse shows that the inverse is also equivariant.

Export E1/v1. Every nonzero G-intertwiner $T : V \to W$ is an equivariant isomorphism.

Acceptance gate. Nonzero T excludes the full kernel and zero image; check both injectivity and surjectivity.

Targeted repair. If only injectivity has been shown, use irreducibility of the codomain to establish surjectivity. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Extract scalar endomorphisms

Object and role. Subtract an eigenvalue, then apply the nonzero-isomorphism alternative. **Local obligation:** $\mathrm{End}_G(V) = k\,\mathrm{id}$ and the stated Hom-space dimensions follow.

Inputs. E1/v1 and a finite-dimensional irreducible V over algebraically closed k. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $\mathrm{End}_G(V) = k\,\mathrm{id}$ and the stated Hom-space dimensions follow.

K — Knowledge

E1/v1 and a finite-dimensional irreducible V over algebraically closed k. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Subtract an eigenvalue, then apply the nonzero-isomorphism alternative.

V — Validation

Eigenvalue existence is precisely where algebraic closure and finite dimension enter.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For $A \in \mathrm{End}_G(V)$ choose an eigenvalue $\lambda \in k$. The map $A - \lambda I$ is equivariant and has a nonzero kernel. E1 says it cannot be a nonzero map, because then it would be invertible. Therefore $A = \lambda I$.

N2.2 If $V \ncong W$, E1 rules out every nonzero intertwiner. If $S : V \to W$ is a fixed isomorphism, every intertwiner T satisfies $S^{-1}T \in \mathrm{End}_G(V)$, so $T = \lambda S$. Thus the Hom-space is one-dimensional in the isomorphic case.

Export E2/v1. $\mathrm{End}_G(V) = k\,\mathrm{id}$ and the stated Hom-space dimensions follow.

Acceptance gate. Eigenvalue existence is precisely where algebraic closure and finite dimension enter.

Targeted repair. If a real irreducible endomorphism is declared scalar, check whether it has a real eigenvalue; retain the algebraically closed field for this conclusion. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Control intertwiners by invariant subspaces

1. **N1.1.** If $v \in \ker T$, then $T\rho_V(g)v = \rho_W(g)Tv = 0$, so the kernel is invariant. The image is invariant because $\rho_W(g)Tv = T\rho_V(g)v$.

2. **N1.2.** Irreducibility gives $\ker T = 0$ or V and $\operatorname{im} T = 0$ or W. Nonzero T forces $\ker T = 0$ and image W. Thus T is a bijective linear map, and multiplying the intertwining equality by its inverse shows that the inverse is also equivariant.

B. N2 — Extract scalar endomorphisms

1. **N2.1.** For $A \in \mathrm{End}_G(V)$ choose an eigenvalue $\lambda \in k$. The map $A - \lambda I$ is equivariant and has a nonzero kernel. E1 says it cannot be a nonzero map, because then it would be invertible. Therefore $A = \lambda I$.

2. **N2.2.** If $V \ncong W$, E1 rules out every nonzero intertwiner. If $S : V \to W$ is a fixed isomorphism, every intertwiner T satisfies $S^{-1}T \in \mathrm{End}_G(V)$, so $T = \lambda S$. Thus the Hom-space is one-dimensional in the isomorphic case.

□

V. WORKED ORCHESTRATION

For the one-dimensional complex characters of C_3, let a generator act by ω on V and by ω^2 on W, where $\omega^3 = 1$ and $\omega \neq 1$. A linear map is multiplication by a, and equivariance gives $a\omega = \omega^2 a$, hence $a = 0$. Each character's endomorphism algebra is $\mathbb{C}$.

A. Boundary and hypothesis audit

Over $\mathbb{R}$, let C_4 act on $\mathbb{R}^2$ by a rotation J through 90 degrees. There is no invariant real line, so the representation is irreducible. Yet J commutes with the action and is not a real scalar matrix. The first part of Schur's lemma holds; the scalar conclusion does not.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 export: every endomorphism of a real irreducible representation is scalar. **Validation witness:** the rotation J. **Diagnosis:** Knowledge imported an eigenvalue over a field where none need exist. **Repair:** apply scalar Schur only over the algebraically closed field in §1. The broader real conclusion is replaced by the weaker statement that nonzero equivariant endomorphisms are invertible. The complex theorem is unchanged.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use the Hom-space dimension in RT-03 to convert an averaging projection into character orthogonality.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.